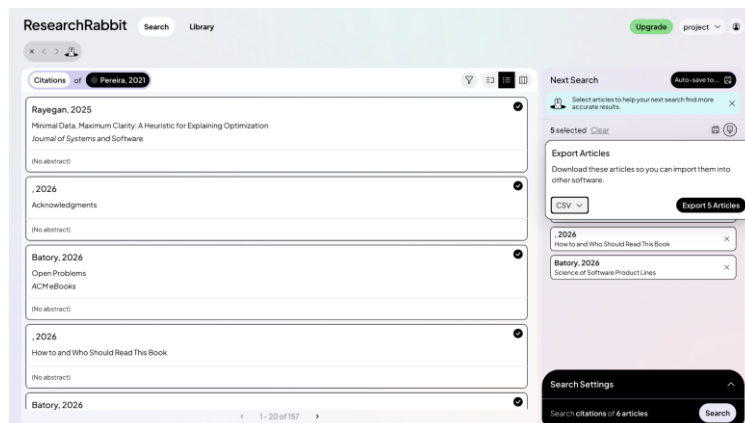
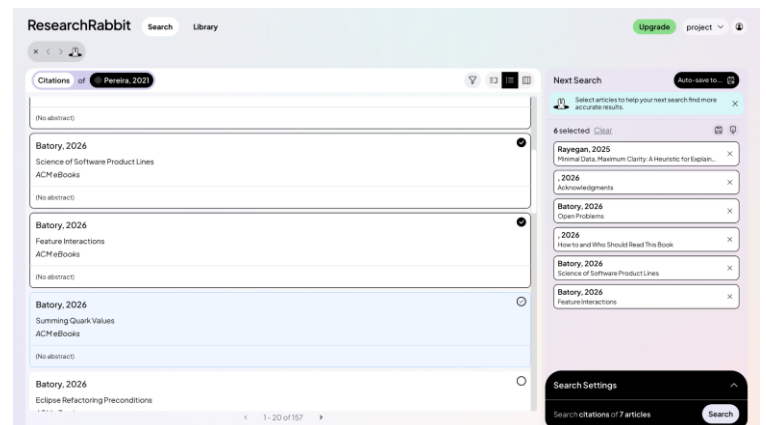
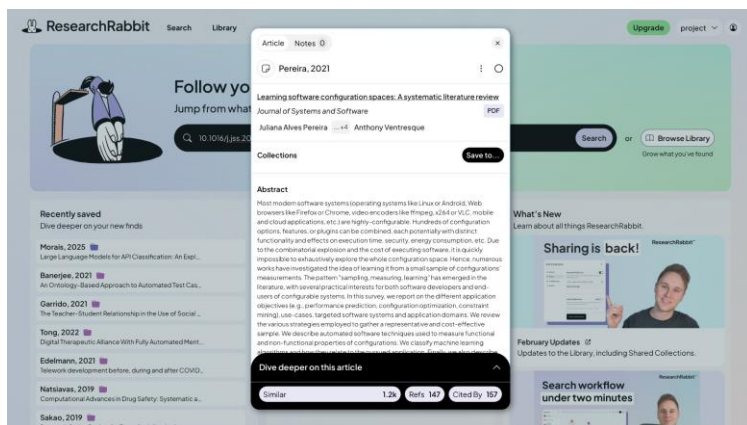
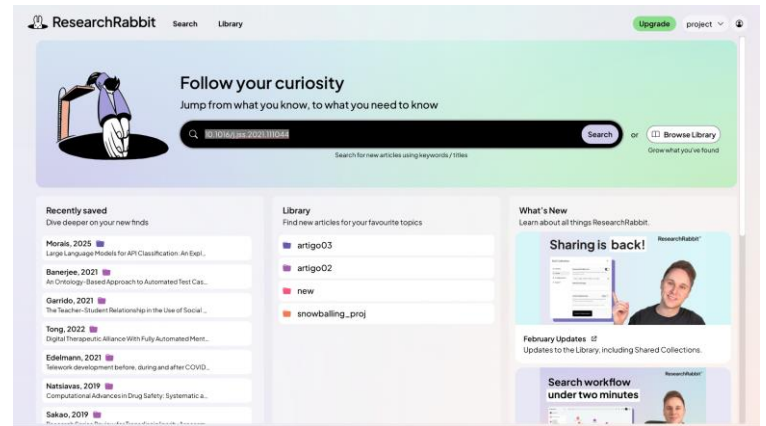
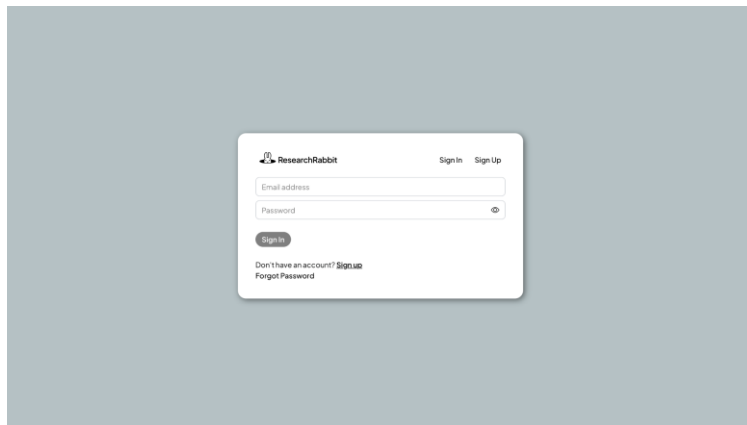


Guide for Exporting Forward Snowballing Results as CSV Files

RESEARCH RABBIT:

For ResearchRabbit, it is first necessary to log in to the platform. The forward snowballing results are then collected directly from ResearchRabbit. For each seed article, the paper is searched using its title or DOI. Once the correct record is found, the Cited by section can be accessed and explored. This section shows the papers that cite the selected seed article, which represent the forward snowballing results returned by ResearchRabbit for that specific seed.

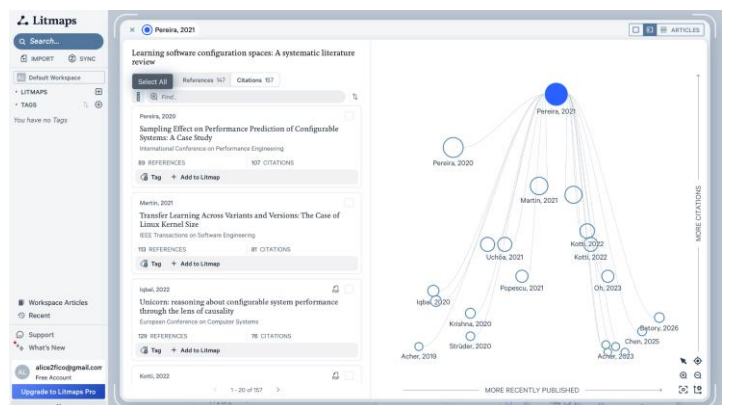
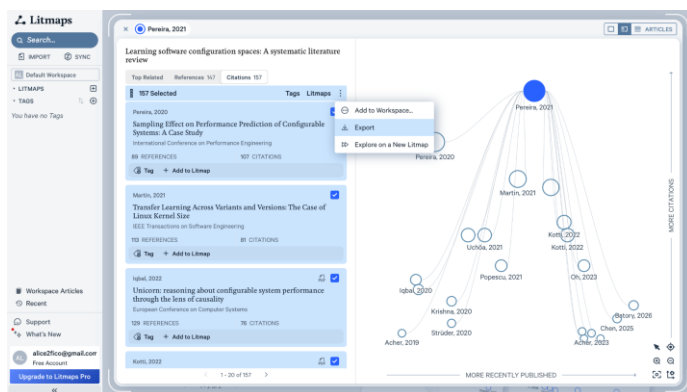
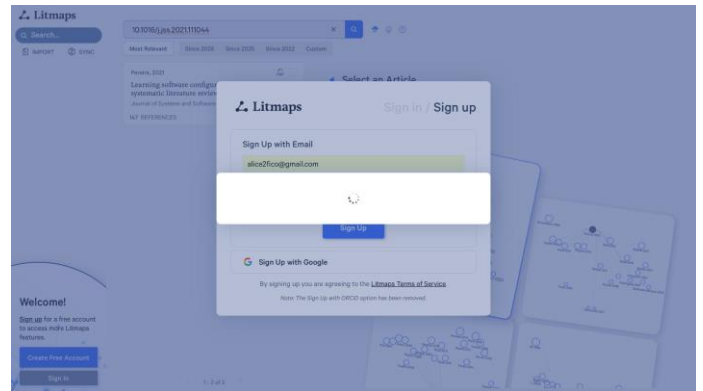
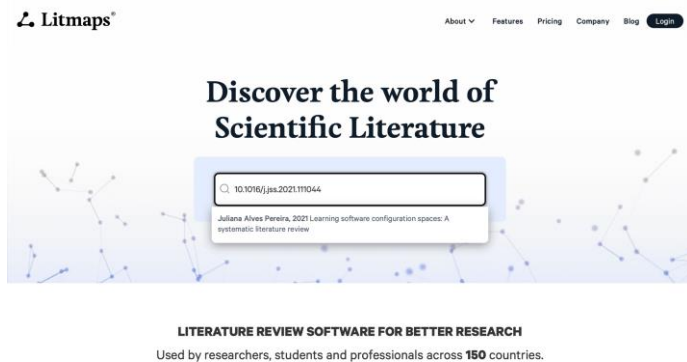
After accessing the list, the results are selected individually and exported from the platform as a CSV file. Each file is then saved according to the naming convention adopted in the study.

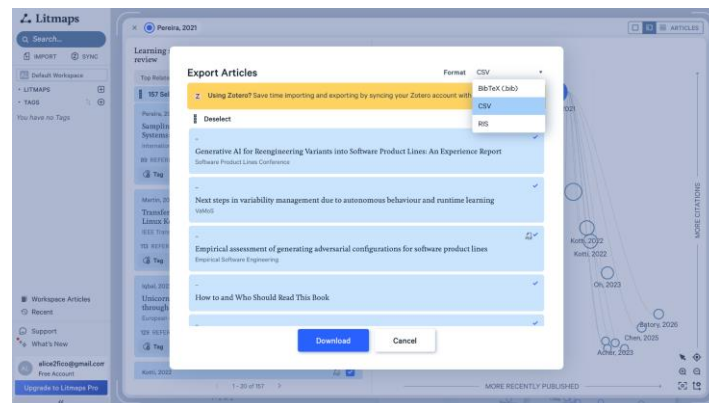


LITMAPS:

For Litmaps, it is first necessary to log in to the platform. Similarly to ResearchRabbit, the forward snowballing results are collected directly from the platform, and login is required to access and explore the Cited by section.

For each seed article, the search is performed by entering the article title or DOI in order to identify the correct paper. Once the seed article is found, the **Cited by** section can be accessed and explored. This section shows the citation relationships associated with the selected article and allows the user to export the results as a CSV file. The papers listed in this section represent the forward snowballing results returned by Litmaps for that specific seed.





SEMANTIC SCHOLAR:

Semantic Scholar does not provide a manual CSV export option from its interface. Instead, the list of citing articles is retrieved through the Semantic Scholar Graph API.

In this case, the seed article is first identified using its DOI, title, or Semantic Scholar paper ID. Then, a Python script sends a request to the API endpoint dedicated to citations. This endpoint returns the papers that cite the selected seed article, which correspond to the forward snowballing results.

The API response is returned in JSON format and is then converted into a structured table using Python. Finally, the table is exported as a CSV file and saved according to the naming convention adopted in the study. The resulting CSV contains the articles that cite the seed article according to Semantic Scholar, together with the metadata available through the API.

```
extraction_semanticScholar-2.py
Apri con Visual Studio Code

import requests
import pandas as pd

# Paper to analyze
paper_id = "10.1145/3701625.3701652"

# === REAL supported fields from your API test ===
fields = [
    "title",
    "abstract",
    "authors",
    "year",
    "venue",
    "journal",
    "url",
    "isOpenAccess",
    "openAccessPdf",
    "citationCount",
    "referenceCount",
    "isInfluential",
    "publicationTypes",
    "fieldsOfStudy",
    "externalIds",
    "influentialCitationCount"
]

fields_param = ",".join(fields)

# URL definition
url = (
    f"https://api.semanticscholar.org/graph/v1/paper/"
    f"{paper_id}/citations?limit=1000&fields={fields_param}"
)

# Check if it works
res = requests.get(url)
print("HTTP status:", res.status_code)

data = res.json()
print("Response keys:", list(data.keys()))

if "data" in data:
    citations = []

    for c in data["data"]:
        p = c.get("citingPaper", {}) or {}
        record = {}

        # copy all supported fields safely
        for f in fields:
            value = p.get(f, None)

            # convert dict fields (like openAccessPdf, externalIds) to string
            if isinstance(value, dict):
                record[f] = str(value)
            else:
                record[f] = value
```

OPEN ALEX:

OpenAlex is another tool used to collect the forward snowballing results for each selected seed article. For each seed article, the search is performed directly on the OpenAlex platform, using the article title or its DOI in order to identify the correct scholarly work (the research through article title gives the option to open the cited by section).

After opening the page corresponding to the seed article, the list of works citing that article can be viewed. These citing works represent the forward snowballing results returned by OpenAlex, since they correspond to the papers that cite the selected seed article.

The results are then exported directly as a CSV file, after the required log in, and saved according to the naming convention adopted in the study. Alternatively, the same type of data can also be retrieved

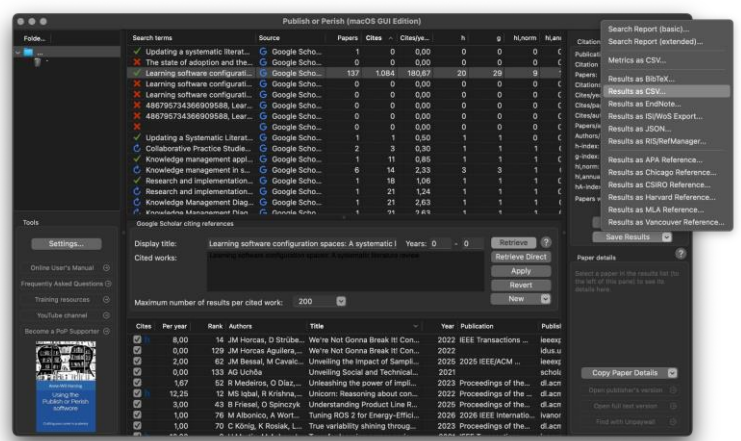
programmatically through the OpenAlex API by querying the works that cite the selected seed article. In this case, the API response can be processed with a Python script and exported as a CSV file.

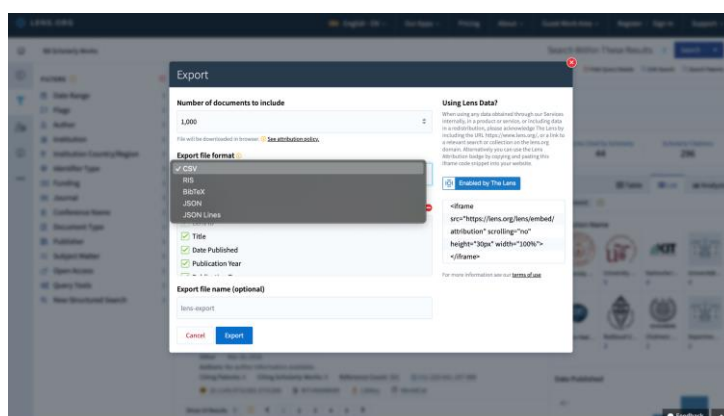
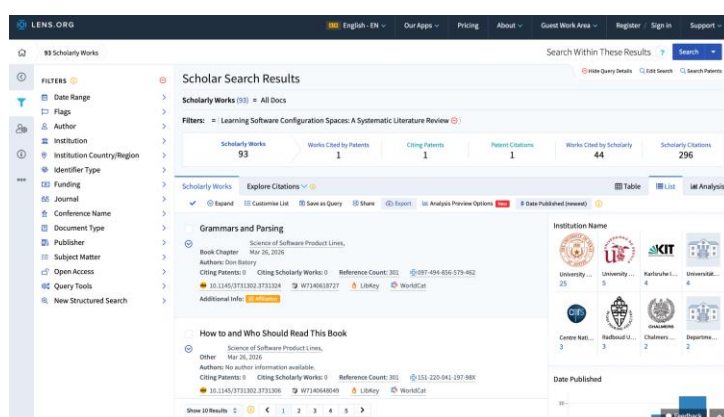
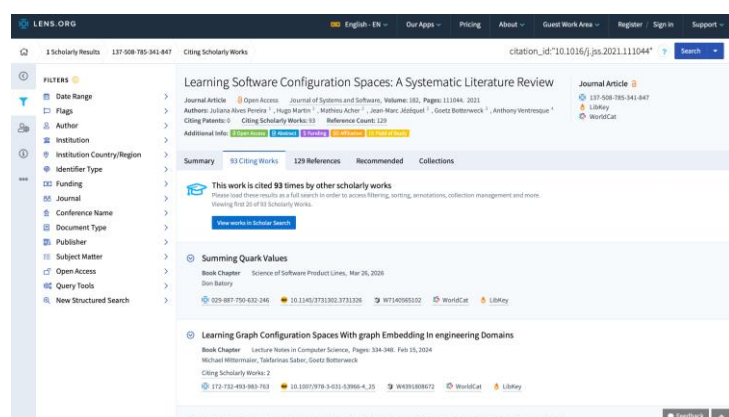
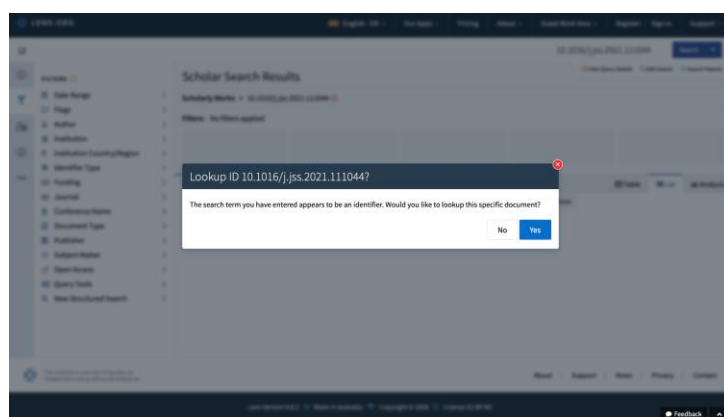
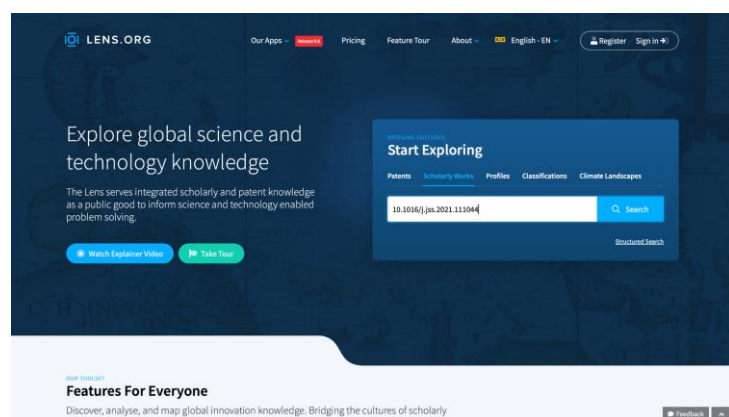
The screenshots illustrate the OpenAlex web interface and its functionality. The top-left screenshot shows the search results for the article 'Learning software configuration spaces: A systematic literature review'. The interface includes filters for Year, Open Access (72.2%), Topic, and Institution. The top-right screenshot shows the same results with a 'Download 103 works' button. The bottom-left screenshot shows a 'Sign in required' dialog box. The bottom-right screenshot shows a 'Welcome back' login screen.

GOOGLE SCHOLAR:

For Google Scholar, the forward snowballing results are collected using an auxiliary platform called **Publish or Perish**, since Google Scholar does not provide a direct option to export citation results as a CSV file from its web interface. Publish or Perish is therefore used as an external tool to search Google Scholar and export the list of citing articles. For each seed article, the search in Publish or Perish is performed by selecting Google Scholar as the data source and entering the title of the seed article. Once the correct paper is identified, it can be selected in order to retrieve the articles citing that seed paper. These citing articles represent the forward snowballing results obtained from Google Scholar. After the results are collected, the list of citing papers can

Once the corresponding article page is found, the list of citing works can be accessed through the citation information provided by the platform. The citing works represent the forward snowballing results for that specific seed article. After accessing this list, the results are exported directly from Lens in CSV format.



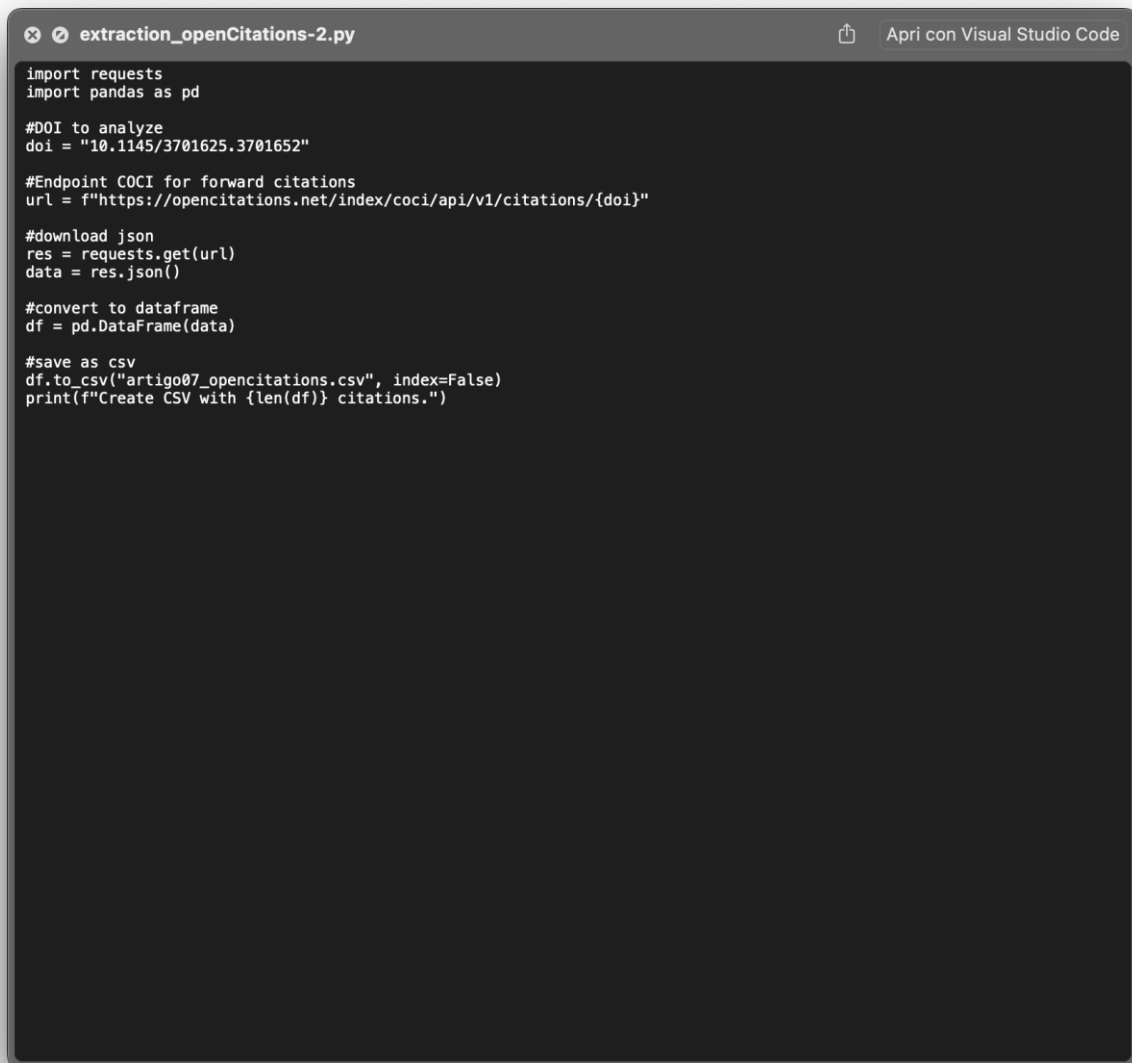


OPEN CITATIONS:

OpenCitations is used through its COCI API, which provides open DOI-to-DOI citation data. Unlike tools that offer a manual CSV export from the web interface, the results from this tool are collected programmatically using a Python script.

For each seed article, the DOI is used as input to query the COCI API and retrieve the works that cite it. These works represent the forward snowballing results. The API response is then converted into a structured table and exported as a CSV file.

The resulting CSV mainly contains citation-level information, the exported data are fewer compared to other tools.

A screenshot of a Visual Studio Code editor window. The title bar at the top shows a file named 'extraction_openCitations-2.py' and a button that says 'Apri con Visual Studio Code'. The editor area contains a Python script that uses the 'requests' and 'pandas' libraries to fetch citation data from an API and save it as a CSV file. The script includes comments in Italian explaining each step: importing libraries, defining the DOI, setting the endpoint, downloading the JSON data, converting it to a DataFrame, and saving it as a CSV. The DOI used is '10.1145/3701625.3701652'. The CSV file is named 'artigo07_opencitations.csv'.

```
import requests
import pandas as pd

#DOI to analyze
doi = "10.1145/3701625.3701652"

#Endpoint COCI for forward citations
url = f"https://opencitations.net/index/coci/api/v1/citations/{doi}"

#download json
res = requests.get(url)
data = res.json()

#convert to dataframe
df = pd.DataFrame(data)

#save as csv
df.to_csv("artigo07_opencitations.csv", index=False)
print(f"Create CSV with {len(df)} citations.")
```

SNOWMAP:

Snowmap is the tool developed within this study to support the automation of forward snowballing. It is specifically designed to retrieve and organize the articles that cite a given seed article by integrating data from multiple scholarly sources.

To collect the forward snowballing results, the user first inserts the information of the seed article, such as its DOI or title, into the Snowmap interface. After starting the search, the tool queries the integrated data sources and retrieves the list of papers that cite the selected article. These citing papers represent the forward snowballing results for that seed.

Once the search is completed, the retrieved articles can be reviewed directly in the tool. The results are then exported as a CSV file and saved according to the naming convention adopted in the study. The exported CSV

contains the set of citing articles identified by Snowmap, including bibliographic metadata collected from the integrated APIs.



Initial AnalysisCriteria Applications



SnowMap

The Academic Curator for deep citation mapping and research discovery

Citations (forward)

References (backward)

Search by DOI or Title

10.1016/j.jse.2021.102494

Search



© 2021 AISL Lab (Artificial Intelligence and Software Engineering)



10.1016/j.jse.2021.102494Search

Initial AnalysisCriteria Applications

Systematic Literature Review

READ MORE

Software, Anthony Vermeirque

system), Software system, Artificial intelligence, Cloud computing, Software engineering, Machine learning, Operating system

Cited By

ReportExclude All

ENTRY	ABSTRACT	AUTHORS	OPEN ACCESS	URL	KEYWORDS	LANGUAGE	PAGES	CITATIONS	ACTIONS
10.1016/j.jse.2021.102494 Semantic-aware OS Tuning Language READ MORE	Online OS tuning can improve long-running services, but existing controllers are poorly matched to...	Georgios Liargovas, Mihai Mitu, Jean-Michel Franke, Kostas Kallias	-	https://doi.org/10.1016/j.datak.2025.102494	-	en	102494	0Citations	✓✕
Knowledge Engineering									
1010 Software Product READ MORE	-	Don S. Bailey	-	https://doi.org/10.1046/j.3793302	-	en	-	0Citations	✓✕
1010.2021.17857 M. Leveraging Modeling to Tune hyperparameters READ MORE	Large Language Models (LLMs) are being increasingly used across a wide range of tasks, however, the...	Nadia Bni, Clement Quinlan, Romain Rouvoy	-	https://doi.org/10.48550/arxiv.2602.17857	-	-	-	0Citations	✓✕



10.1016/j.jse.2021.102494Search

Initial AnalysisCriteria Applications

Seed Paper Analysis

RESEARCH IDENTITY	ABSTRACT	AUTHORS	OPEN ACCESS	URL	KEYWORDS	LANGUAGE	PAGES	CITATIONS
10.1016/j.jse.2021.102494 Learning Software Configuration Spaces: A Systematic Literature Review READ MORE	Most modern software systems (operating systems like Linux or Android, web browsers like Firefox, ...)	Juliana Alves Pereira, Hugo Martin, Mathieu Achari, Jean-Michel Franke, Anthony Vermeirque	-	https://iris.hal.science/iris-02348791	Computer science, Plug-in, Software, Android (operating system), Software system, Artificial intelligence, Cloud computing, Software engineering, Machine learning, Operating system	en	-	150Citations

Cited By

ReportExclude All

RESEARCH IDENTITY	ABSTRACT	AUTHORS	OPEN ACCESS	URL	KEYWORDS	LANGUAGE	PAGES	CITATIONS
10.1016/j.jse.2021.102494 Semantic-aware OS Tuning with Large Language Models READ MORE	Online OS tuning can improve long-running services, but existing controllers are poorly matched to...	Georgios Liargovas, Mihai Mitu, Jean-Michel Franke, Kostas Kallias	-	https://doi.org/10.1016/j.datak.2025.102494	-	en	102494	0Citations

2025 DataCamp Knowledge Engineering